

Active Inference

Principles and Applications

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Active Inference

Definition (Active Inference)

A system is conducting active inference if it adjusts its **beliefs** about the environment, as adjusting the environment so as to minimise the **surprise** of observations generated from its environment.

Outline

1 High Road

2 Low Road

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2 Low Road

Dynamical System

- Physical State Space X

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- Allow for the representation $X = S \times R$

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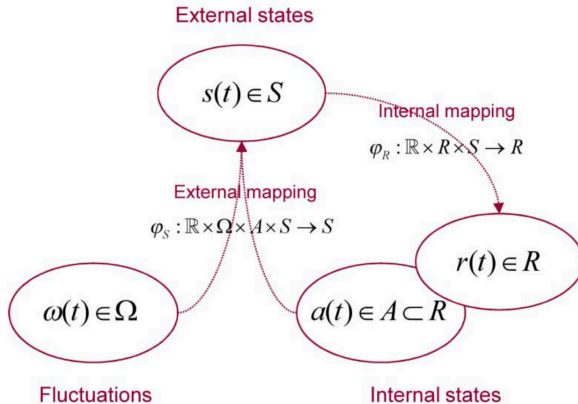


Figure: Friston (2012)

Ergodic Density

Definition (Ergodic Density)

For $x \in X$. We define the ergodic density p as:

$$p(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta(x - x(t)) dt$$

Definition (Active System)

We consider a biological system active if it possess an internal map φ_R^*

$$\varphi_R^* = \operatorname{argmin}_{\varphi_R} H(S) = - \int p(s) \ln p(s) \, ds$$

Minimisation of Surprise

Theorem (Principle of Least Action)

The internal states of an active system minimise surprise, $\mathcal{L}(s(t)) = -\ln p(s(t))$.

$$\varphi_R^*(t) = \operatorname{argmin}_r \mathcal{L}(s(t))$$

1 High Road

2 Low Road

Perception as unconscious inference

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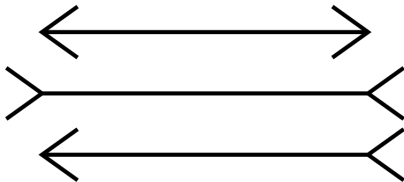


Figure: Howe and Purves (2005)

Perception as unconscious inference

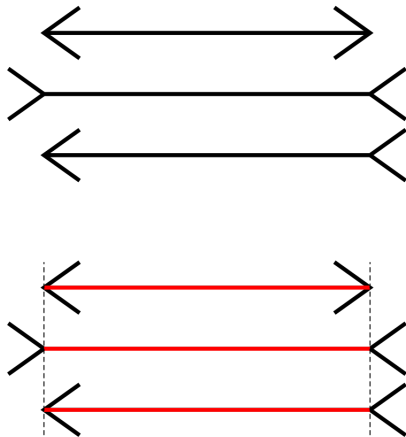


Figure: Howe and Purves (2005)

Bayesian Inference

- Generative Model: $p(x, y)$

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- Bayes' Rule:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

Variational Inference

$$p(y) = \int_{\mathcal{X}} p(x, y) \, dx \quad (1)$$

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Definition (Variational Free Energy)

$$\mathcal{F}(q, y) = \int q(x) \ln \frac{q(x)}{p(x, y)} \, dx$$

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Definition (Variational Free Energy)

$$\mathcal{F}(q, y) = \int q(x) \ln \frac{q(x)}{p(x, y)} \, dx$$

$$\mathcal{F}(q, y) = D_{KL} [q(x) \| p(x)] - \mathbb{E}_q [\ln p(y|x)]$$

Surprise Minimisation

$$\mathcal{F}(q, y) = D_{KL}[q(x) \| p(x|y)] - \ln p(y)$$

Surprise Minimisation

$$\mathcal{F}(q, y) = D_{KL} [q(x) \| p(x|y)] - \ln p(y) \quad (2)$$

$$\geq \underbrace{-\ln p(y)}_{\text{Surprise}} \quad (3)$$

References

- Friston, K. (2012). A free energy principle for biological systems. *Entropy*, 14(11):2100–2121.
- Howe, C. Q. and Purves, D. (2005). The müller-lyer illusion explained by the statistics of image–source relationships. *Proceedings of the National Academy of Sciences*, 102(4):1234–1239.